CUSTOM BOARD — COMPONENT LIST & BUILD GUIDE

RoboFace — Custom ESP32 Board

Your own PCB built around the bare **ESP32-WROOM-32E** module (no dev board). Drives a 0.96" SSD1306 OLED, syncs to Firebase over WiFi, and updates over the air.

~17 unique parts (line items)	**~30** total pieces to place	**4 MB+** flash needed (for OTA)

> **Why the module, not a bare chip?** The ESP32-WROOM-32E already contains the ESP32 chip, 4 MB flash, 40 MHz crystal and a tuned PCB antenna — pre-certified. It removes all RF design risk. A truly bare-chip design is covered briefly on the last page.

Block diagram

```
              +5V                 3.3V rail
USB-C ──► ESD ──► [ LDO RT9013-3.3 ] ─┬───────────────► ESP32-WROOM-32E ─┐
  │  D+/D-                             └──► OLED VCC         │  EN / IO0   │ I2C
  ▼                                                          ▲            │ SDA=GPIO21
[ CP2102N / CH340C USB-UART ] ── RX/TX ───────────────────►  │            │ SCL=GPIO22
  │  DTR + RTS                                                            ▼
  └──► [ Q1 / Q2 auto-reset ] ──► EN & IO0              [ SSD1306 128x64 OLED ]
```

USB-C gives 5 V (power) and data (flashing). The LDO drops 5 V → 3.3 V for the ESP32 and OLED. The USB-UART bridge programs the chip; its DTR/RTS lines drive a two-transistor circuit that auto-resets the ESP32 into download mode. I^2C (GPIO21/22) talks to the OLED.

Bill of materials

REF	COMPONENT	VALUE / PART	QTY	PURPOSE
Core				
U1	ESP32 module	ESP32-WROOM-32E-N4 (4 MB)	1	MCU + WiFi + flash + antenna
DS1	OLED display	0.96" SSD1306 128×64, I²C (0x3C)	1	The robot face
Power — USB 5 V → 3.3 V				
J1	USB-C receptacle	16-pin SMD (or Micro-USB)	1	Power + data in
R1,R2	Resistor	5.1 kΩ	2	USB-C CC1/CC2 pulldown
U2	LDO regulator	RT9013-33 (low-dropout, 500 mA) / AMS1117-3.3 if USB-only	1	3.3 V rail
C1,C2	Ceramic cap	10 µF / 16 V	2	LDO input & output
C3	Bulk cap	22–100 µF	1	Rides out WiFi current spikes
C4–C8	Ceramic cap	100 nF (0402/0603)	5	Per-pin decoupling
Programming — USB-UART + auto-reset				
U3	USB-UART bridge	CP2102N-A02 (or CH340C)	1	Flash + serial monitor
Q1,Q2	NPN transistor	MMBT3904 (SOT-23)	2	DTR/RTS → auto reset to boot
R3–R6	Resistor	10 kΩ	4	EN & IO0 pull-ups + reset net
C9	Ceramic cap	1 µF	1	EN power-on reset delay
D1	ESD array (optional)	USBLC6-2SC6	1	Protect USB D+/D−
Buttons, I²C & indicator				
SW1	Tactile switch	6×3.5 mm SMD	1	EN / Reset
SW2	Tactile switch	6×3.5 mm SMD	1	BOOT (IO0) for flashing
R7,R8	Resistor	4.7 kΩ	2	SDA/SCL pull-ups (skip if OLED has them)
D2,R9	LED + resistor	0603 LED + 1 kΩ	2	Power indicator
J2	OLED connector	4-pin 2.54 mm header / JST-SH	1	Mount/connect the panel

Key connections (ESP32-WROOM-32E)

MODULE PIN	CONNECTS TO	WHY
3V3	3.3 V rail + C4 (100 nF) + C3 bulk	Power
GND	Ground plane	Power
EN	10 kΩ → 3V3, 1 µF → GND, SW1, Q-reset	Enable / reset
IO0	10 kΩ → 3V3, SW2, Q-reset	Boot select (download mode)
TXD0 (GPIO1)	USB-UART RXD	Programming

RXD0 (GPIO3)	USB-UART TXD	Programming
GPIO21	OLED SDA	I^{2}C data
GPIO22	OLED SCL	I^{2}C clock

Assembly & layout notes

- **Flash size:** use the **-N4 (4 MB)** or larger WROOM — OTA stores two app copies (your build uses the 1.9 MB "Minimal SPIFFS" partition).
- **Antenna keep-out:** the module hangs over the PCB edge; keep *all copper, traces and ground pour out* of the area under the antenna, or WiFi range drops badly.
- **Decoupling:** a 100 nF cap as close as possible to the module's 3V3 pin, plus the 22–100 µF bulk on the 3.3 V rail — this prevents the random resets WiFi current spikes cause.
- **I^2C pull-ups:** need ~4.7 kΩ on SDA & SCL. Most OLED modules already include them — if yours does, leave R7/R8 off (parallel resistors over-load the bus).
- **Auto-reset:** CP2102/CH340 `DTR` + `RTS` through Q1/Q2 toggle `EN` and `IO0` so the toolchain flashes without pressing buttons.
- **Strapping pins:** don't tie GPIO0/2/5/12/15 to anything that's driven at power-up. You only use GPIO21/22 → safe.
- **First flash:** over USB once (download `roboface_full.bin` from your GitHub release, or use the Arduino "Minimal SPIFFS" partition). After that, every update is OTA — no cable.
- **Power option:** for battery, add a 3.7 V LiPo + TP4056 charger and keep the *low-dropout* LDO (RT9013) so it doesn't brown out as the cell drains.

If you really want a bare-chip design (no module)

Replacing the WROOM module means adding everything it integrates. Only attempt this if you're comfortable with RF layout.

EXTRA PART	VALUE / PART	QTY	WHY THE MODULE ALREADY HAS IT
ESP32 chip	ESP32-D0WDQ6-V3 (QFN-48)	1	The SoC itself
SPI flash	GD25Q32 (4 MB, SOP-8)	1	Program storage
Crystal	40 MHz + 2× ~12 pF caps	1	System clock
RF match	π-network: inductor + 2 caps (tuned)	1	Antenna impedance match (50 Ω)
Antenna	Chip antenna or u.FL + 50 Ω trace	1	WiFi radiation
Extra decoupling	100 nF × several + 10 µF	—	Stable VDD rails

> **Recommendation:** stick with the WROOM module. The bare chip saves only a little cost but adds RF tuning, impedance-controlled traces and FCC/CE re-certification.